

YouTube Study Resources — AI/ML Engineer & Python Developer Interview Prep

Pairs with: AI-ML-Interview-Prep-Guide.md

Suggested order to watch these: **ML fundamentals** → **LLMs/Transformers** → **RAG** → **Python internals** → **System design/MLOps** (last, right before interviews).

1. LLM Fundamentals

Topic	Video / Playlist	Link
Big-picture intro to LLMs (pretraining, chatbots, transformers)	3Blue1Brown — "Large Language Models explained briefly"	https://www.youtube.com/watch?v=LPZh9BOjkQs
Deep dive: how transformers + attention actually work (visual, best explanation available)	3Blue1Brown — Neural Networks / Deep Learning playlist (includes the Attention chapter)	https://www.youtube.com/playlist?list=PLZZWrBYkx7Otcjr3eCLZDCgfpqnxMY29s
Quick landscape of current models (GPT, Claude, Gemini, Llama, etc.)	"Every Large Language Model Explained in 17 Minutes"	https://www.youtube.com/watch?v=9UpLmjIDgPg

What to focus on: pretraining vs instruction tuning vs RLHF, tokens, context window, embeddings, temperature, hallucination — all covered conceptually in the 3Blue1Brown series.

2. RAG (Retrieval Augmented Generation)

Topic	Video / Playlist	Link
Full hands-on RAG build (matches your DocuMind stack: LangChain, ChromaDB)	Krish Naik — “Complete RAG Crash Course With LangChain in 2 Hours”	https://www.youtube.com/watch?v=o126p1QN_RI
More RAG variants: Agentic RAG, LangGraph, vector DB deep-dives	Krish Naik — full channel playlists	https://www.youtube.com/@krishnaik06/playlists

What to focus on: chunking strategy, embeddings, vector search, retrieval + generation pipeline, and how to explain RAGAS metrics (faithfulness/relevancy) — since you already have real numbers from DocuMind to quote.

3. System Design & Production AI

Topic	Video / Playlist	Link
ML/AI system design interview style (scale, tradeoffs, structured answers)	ByteByteGo channel	https://www.youtube.com/ByteByteGo
Deployment, monitoring, drift, versioning, MLOps pipeline	“MLOps, Machine Learning System Design” playlist	https://www.youtube.com/playlist?list=PLLvYm28hIDqI0zT8fzGzJwd2dnEccYhSG

What to focus on: practice structuring answers as: clarify scale → propose architecture → call out bottlenecks → monitoring/reliability. Also good to mentally redesign DocuMind for 10,000 concurrent users using these ideas.

4. ML & Deep Learning Fundamentals

Topic	Video / Playlist	Link
Neural networks,		

backpropagation, gradient descent, bias/variance, cross-validation (all in bite-sized, “clearly explained” videos)	StatQuest with Josh Starmer — Neural Networks / ML Fundamentals playlist	https://www.youtube.com/playlist?list=PLblh5JKOoLUlcdlgu78MnlATeyx4cEVeR
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What to search within the channel: “Gradient Descent, Step-by-Step,” “Machine Learning Fundamentals: Bias and Variance,” “Machine Learning Fundamentals: Cross Validation,” “Neural Networks Pt. 2: Backpropagation Main Ideas” — each is 10–20 min and very interview-friendly.

5. Python Developer Topics

Topic	Video / Playlist	Link
OOP fundamentals: classes, <code>@classmethod</code> vs <code>@staticmethod</code> , inheritance, dunder methods, property decorators	Corey Schafer — Python OOP Tutorials playlist	https://youtube.com/playlist?list=PL-osiE80TeTsqhluOgKhwlXsIBIdSeYtc
Full channel (decorators, generators, GIL-adjacent topics, general Python)	Corey Schafer — all playlists	https://www.youtube.com/user/schafer5/playlists
async/await and asyncio explained (relevant since you use FastAPI)	“Python Asynchronous Programming Tutorial: Asyncio, async & await Explained”	https://www.youtube.com/watch?v=ngXbyui-weA

What to focus on: the GIL’s effect on threading vs multiprocessing vs asyncio (know when to use which), decorators (write one live if asked), generators/ `yield`, and how FastAPI’s async routes actually behave under the hood.

Suggested Weekly Plan

- **Week 1:** StatQuest ML fundamentals + Corey Schafer Python OOP/decorators
- **Week 2:** 3Blue1Brown LLM/transformer series + asyncio video
- **Week 3:** Krish Naik RAG crash course (rebuild a mini version yourself alongside DocuMind)
- **Week 4:** ByteByteGo + MLOps playlist, then mock-answer the system design questions from your prep guide out loud